预配计算机以便进行驱动程序部署和测试 (WDK 10)

2024/09/27

*预配目标计算机或测试计算机*是配置计算机以自动部署、测试和调试驱动程序的过程。 要预配计算机，请使用 Microsoft Visual Studio。

一个测试和调试环境具有两台计算机：主计算机和目标计算机。 目标计算机也称为“测试计算机”。 在主计算机上的 Visual Studio 中开发和构建驱动程序。 调试程序在主机上运行并且位于 Visual Studio 用户界面中。 当测试和调试驱动程序时，驱动程序在目标计算机上运行。

确保主机和目标计算机可以通过名称互相 ping。 如果两台电脑都属于同一个工作组或网络域，那么这一过程就比较简单。 如果计算机位于工作组中，请使用路由器而不是集线器或交换机连接它们。

为部署驱动程序调配系统时，主机和目标计算机必须使用相同版本的 Windows 操作系统。

> **提示**
>
> 有关 WDK 上的社区支持，请参阅 Microsoft Q&A 上的 **Windows 硬件性能论坛**。

准备要预配的目标计算机

1. 在目标计算机上，安装将用于运行和测试驱动程序的操作系统。
2. 安装 WDK。 只有计划在目标计算机上进行驱动程序开发时，才需要安装 Visual Studio。
3. 如果目标计算机启用了安全启动，请将其禁用。 有关统一可扩展固件接口 (UEFI) 和安全启动的信息，请参阅 UEFI 固件。
4. 如果目标计算机是 VM，则应在非增强模式下启动它，以便禁用 VM 增强模式配置，请参阅与虚拟机共享设备
5. 如果要在目标计算机上部署测试驱动程序，请使用 **bcdedit /set testsigning** 从提升的命令启用测试签名。

6. 如果要使用 WDTF 测试驱动程序，请确保使用属性页启用驱动程序验证程序，并在部署后重启目标计算机

    - 在主计算机上，在 Visual Studio 中右键单击驱动程序项目，前往**属性** > **配置属性** > **驱动程序安装** > **驱动程序验证** > **启用驱动程序验证**

7. 在目标计算机上，运行与目标计算机平台匹配的 WDK 测试目标设置 MSI。 可以在 Remote 下的 Windows 驱动程序工具包 (WDK) 安装目录中找到该 MSI。

    示例：C:\Program Files (x86)\Windows Kits\10\Remote\x64\WDK Test Target Setup x64-x64_en-us.msi

8. 如果目标计算机运行的是 Windows Server，请查找刚才通过 WDK 测试目标设置 MSI 创建的 DriverTest 文件夹。 （示例：c:\DriverTest）。 选择并按住（或右键选择）**DriverTest** 文件夹，然后选择**属性**。 在“安全性”选项卡上，向“经过身份验证的用户”组授予“修改”权限。

验证主计算机和目标计算机是否可以彼此 ping 通。 打开命令提示符窗口，输入“**ping** ComputerName”。

如果主计算机和目标计算机已加入到一个工作组，但位于不同的子网上，则可能必须调整某些防火墙设置，以便主计算机和目标计算机可以通信。 执行以下步骤：

1. 在目标计算机上的“控制面板”中，转到“网络和 Internet”>“网络共享中心”。 请注意活动网络。 这将是**公用网络**、**专用网络**或**域**。
2. 在目标计算机上的“控制面板”中，转到“系统和安全”>“Windows 防火墙”>“高级设置”>“入站规则”。
3. 在入站规则列表中，查找用于活动网络的所有网络发现规则。 （例如，查找**配置文件**为**专用**的所有网络发现规则。）双击每个规则并打开**作用域**选项卡。在**远程 IP 地址**下选择**任意 IP 地址**。
4. 在入站规则列表中，查找用于活动网络的所有“文件和打印机共享”规则。 对于其中每个规则，双击该规则，然后打开**作用域**选项卡。在**远程 IP 地址**下选择**任意 IP 地址**。

预配目标计算机

现在，可以随时在 Visual Studio 中通过主计算机预配目标计算机。

1. 在主机计算机上的 Visual Studio 中，选择“扩展”菜单，依次指向“驱动程序”和“测试”，然后选择“配置设备”。

2. 在“配置设备”对话框中，选择“添加新设备”。

3. 对于**网络主机名**，输入目标计算机的名称或本地 IP 地址。 选择“预配设备并选择调试程序设置”。

Device Configuration

Adding a new target device

Display name:

MyTestComputer

Network host name:

Provisioning Options

- **Provision device and choose debugger settings**
  Installs and configures components on the target to allow deploying drivers and running tests from Visual Studio. The specified debug settings will also be set on the target, and the connection validated.
- **Manually configure debuggers and do not provision**
  Sets debuggers settings in Visual Studio to connect to the target, but does no validation or provisioning.

Show help

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4. 选择**下一步**。

5. 选择某种调试连接类型，并输入所需的参数。

   有关通过各种类型的连接设置调试的详细信息，请参阅 CHM 中的手动设置 KDNET 网络内核调试或 Windows 调试工具的相关文档。

6. 预配过程将需要几分钟时间，并且可能会自动重新启动目标计算机一到两次。 预配完成后，选择“完成”。

7. **重启计算机**。

另请参阅

将驱动程序部署到测试计算机

预配计算机时 (WDK 8.1)

Provision a computer for driver deployment and testing (WDK 10)

09/27/2024

Provisioning a target or test computer is the process of configuring a computer for automatic driver deployment, testing, and debugging. To provision a computer, use Microsoft Visual Studio.

A testing and debugging environment has two computers: the *host computer* and the *target computer*. The target computer is also called the *test computer*. You develop and build your driver in Visual Studio on the host computer. The debugger runs on the host computer and is available in the Visual Studio user interface. When you test and debug a driver, the driver runs on the target computer.

Make sure that the host and target computers can ping each other by name. This process is simpler if both computers belong to the same workgroup or network domain. If your computers are in a workgroup, connect them with a router rather than a hub or switch.

When provisioning systems for driver deployment, the host and the target machines must have the same version of Windows operating system.

> **Tip**
>
> For community support on the WDK, see the **Windows Hardware Performance Forum** on Microsoft Q&A.

Prepare the target computer for provisioning

1. On the target computer, install the operating system that you'll use to run and test your driver.
2. Install the WDK. You only need to install Visual Studio if you plan on doing driver development on the target computer.

3. If Secure Boot is enabled on the target computer, disable it. For information about Unified Extensible Firmware Interface (UEFI) and Secure Boot, see UEFI Firmware.

4. If the target computer is a VM, it should be started in non-enhanced mode, for disabling VM enhanced mode configuration, see share devices with your virtual machine

5. If you will be deploying a test driver on the target machine, enable test signing from an elevated command with **bcdedit /set testsigning on**

6. If you will be testing the driver with WDTF, make sure to enable Driver Verifier using the properties page and restart the target machine after the deployment

    - On the host computer, in Visual Studio, right click on the driver project go to **Properties** > **Configuration Properties** > **Driver Install** > **Driver Verification** > **Enable Driver Verification**

7. On the target computer, run the WDK Test Target Setup MSI that matches the platform of the target computer. You can find the MSI in the Windows Driver Kit (WDK) installation directory under Remote.

    Example: C:\Program Files (x86)\Windows Kits\10\Remote\x64\WDK Test Target Setup x64-x64_en-us.msi

8. If your target computer is running Windows Server, find the DriverTest folder that was just created by WDK Test Target Setup MSI. (Example: c:\DriverTest). Select and hold (or right-select) the **DriverTest** folder, and choose **Properties**. On the **Security** tab, give **Modify** permission to the **Authenticated Users** group.

Verify that the host and target computers can ping each other. Open a Command Prompt window, and enter **ping** *ComputerName*.

If the host and target computers are joined to a workgroup and are on different subnets, you might have to adjust some firewall settings so that the host and target computers can communicate. Follow these steps:

1. On the target computer, in Control Panel, go to **Network and Internet** > **Network Sharing Center**. Note your active network. This will be **Public network**, **Private network**, or **Domain**.

2. On the target computer, in Control Panel, go to **System and Security** > **Windows Firewall** > **Advanced settings** > **Inbound Rules**.
3. In the list of inbound rules, find all Network Discovery rules for your active network. (For example, find all Network Discovery rules that have a **Profile** of **Private**.) Double click each rule and open the **Scope** tab. Under **Remote IP address**, select **Any IP address**.
4. In the list of inbound rules, locate all File and Printer Sharing rules for your active network. For each of those rules, double click the rule, and open the **Scope** tab. Under **Remote IP address**, select **Any IP address**.

Provision the target computer

Now you're ready to provision the target computer from the host computer in Visual Studio.

1. On the host computer, in Visual Studio, select the **Extensions** menu, point to **Driver**, point to **Test**, and select **Configure Devices**.

2. In the **Configure Devices** dialog, select **Add new device**.

3. For **Network host name**, enter the name or local IP address of your target computer. Select **Provision device and choose debugger settings**.

Device Configuration

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Show help

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4. Select **Next**.

5. Select a type of debugging connection, and enter the required parameters.

   For more information about setting up debugging over various types of connections, see Setting Up KDNET Network Kernel Debugging Manually and the related documentation for the Debugging Tools for Windows.

6. The provisioning process takes several minutes and might automatically reboot the target computer once or twice. When provisioning is complete, select **Finish**.

7. **Restart** the computer.

See Also

Deploying a Driver to a Test Computer

When you provision a computer (WDK 8.1)

Provision a computer for driver deployment and testing (WDK 10)